Mughesh Kumar N R

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Education

University of Virginia *Aug 2025 – Dec 2026*
Master of Science in Computer Science Charlottesville, VA
- **Coursework:** Autonomous AI Agents, LLM Systems & Tool Use, Distributed Systems, Knowledge Graphs.
- Graduate Teaching Assistant: Mentored and instructed 50+ students in hands-on labs, breaking down complex distributed systems, API architectures, and debugging workflows into accessible concepts.
- Judged collegiate hackathons and co-led engineering workshops on production AI reliability and evaluation frameworks.

University of California, Berkeley *Jan 2025 – May 2025*
Startup Semester Program Berkeley, CA
- Completed intensive immersion in Product Management, Enterprise Customer Discovery, and AI Go-to-Market Strategy, translating complex operational workflows into technical platform requirements.
- Awarded **$10,950** UC Berkeley Chancellor's Scholarship for Startup Semester, selected from 600+ international applicants.

SRM Institute of Science and Technology *Aug 2021 – Dec 2024*
Bachelor of Technology in Computer Science and Engineering (AI & ML) Chennai, India
- Awarded University Gold Medal for Academic Excellence; elected Student Body President representing 20,000+ peers, partnering with executive leadership to deploy university-wide digital initiatives.
- Won two international SDG hackathons and **$1,500** in prizes, developing predictive prototypes for industry juries.
- Published two peer-reviewed IEEE research papers and granted an Indian patent for data-driven IoT telemetry systems.

Work Experience

University of Virginia Information Technology Services (ITS) *June 2026 – Aug 2026*
Enterprise Systems Deployment Intern Charlottesville, VA
- Partnered with enterprise IT specialists on a **$6M+ Infrastructure Modernization** project, analyzing telemetry from 6,656 network devices to optimize performance, resource allocation, and operational reliability.
- Built automated **Python** and **REST API** data integration pipelines to extract, clean, and structure unstructured system logs, reducing data preparation latency and diagnostic turnaround by 35%.
- Engineered geospatial telemetry dashboards in ArcGIS and Python, translating complex operational metrics into actionable production visibility and documented data products for university leadership.
- Implemented automated validation scripts and error-handling routines to continuously monitor system health, establishing front-line feedback loops that improved overall deployment reliability.

IBM *June 2023 – July 2023*
Software & AI Engineering Intern India
- Engineered automated end-to-end data pipelines in **Python** and **SQL** across 100,000+ records of structured and unstructured health data, structuring knowledge models with automated validation checks.
- Developed and evaluated Scikit-learn predictive models, achieving 99.4% R^2 accuracy and outperforming baseline models.
- Synthesized statistical findings and workflow insights into documented data products, interactive Tableau dashboards, and technical documentation, communicating trade-offs directly to business partners.

Projects

FaceMark AI — Biometric Feature Extraction & Autonomous Workflow Platform [Live Demo]
Tech Stack: *Python, JavaScript (ES6+), WebRTC, Canvas API, Web Speech API, MTCNN, REST APIs, Vercel.*
- Architected a browser-based biometric identity and verification platform generating 128-D vector embeddings for high-dimensional feature representation and real-time similarity matching.
- Designed automated multi-step verification workflows with Web Speech voice prompts, 30-minute cooldown state machines, and local storage persistence with CSV analytical export.
- Deployed live production web app on Vercel with zero external server dependencies, validating sub-second inference latency, edge compute constraints, and privacy-first data guarantees.

Agri-Vision: Distributed Computer Vision & IoT Diagnostic Pipeline
Tech Stack: *Python, TensorFlow, OpenCV, Microservices, IoT Sensors, REST APIs, NVIDIA Jetson Nano.*
- Designed and deployed an end-to-end computer vision diagnostic pipeline across regional farms, detecting 8+ agricultural anomalies with 96% diagnostic accuracy for field operations.
- Integrated Python microservices with heterogeneous IoT sensor streams and edge hardware (Jetson Nano), structuring real-time telemetry into unified diagnostic schemas.
- Evaluated system performance across latency, compute cost, and accuracy, documenting technical trade-offs, diagnostic evaluations, and operational runbooks for non-technical users.

Technical Skills

AI Agents & LLM Systems: Autonomous Agents, Tool Calling, Knowledge Graphs, RAG, Vector Embeddings, Prompt Tuning.
Languages & Frameworks: Python (FastAPI, PyTorch, Pandas, Scikit-learn), TypeScript, JavaScript, SQL, C/C++.
Systems & Deployment: Microservices, Docker, Linux/Unix, Git, CI/CD, Production Monitoring, Distributed Systems, AWS.
Client-Facing & Product: Forward Deployed Engineering, Customer Discovery, Requirements Gathering, Runbooks.

Publications & Patents

- Energy Efficiency Optimization in Smart Street Lighting Systems via LoRaWAN IoT – IEEE [Publication]
- Handwritten Digit Recognition Comparative Analysis with CNN & LeNet-5 Models – IEEE [Publication]
- IoT-Based Attendance and Monitoring System for Remote Operations Sites – IP India Granted [Patent]